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Traffic Sign Classification using Deep Learning and Computer Vision Techniques

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ABSTRACT

This research presents an enhanced deep learning framework for traffic sign classification using a modified ResNet architecture. The study utilizes selected classes (0–8 and 33–40) from the German Traffic Sign Recognition Benchmark (GTSRB) dataset, encompassing critical traffic indicators such as speed limits and directional signs.

To overcome challenges such as illumination variations, occlusions, and class imbalance, the proposed framework integrates advanced computer vision and deep learning techniques. It utilizes transfer learning with an ImageNet-pretrained ResNet18, modified with a 3×3 convolutional kernel (optimized for 32×32 images), selective layer freezing to retain low-level features, and a dropout-regularized classifier head ($512\rightarrow 256\rightarrow 128\rightarrow 17$) with batch normalization for improved generalization and robustness.

The training methodology incorporates a comprehensive data augmentation pipeline to enhance model robustness and generalization capability. Augmentation techniques include: geometric transformations (random rotation $\pm 15^\circ$, random affine transformations with translation range $\pm 10\%$ and scale variance 0.9-1.1, random perspective distortion with 10% distortion scale); photometric augmentations (color jittering with brightness $\pm 30\%$, contrast $\pm 30\%$, saturation $\pm 20\%$); and cutout-based random erasing (10% probability, 2-15% scale) to simulate occlusions.

Experimental evaluations demonstrate that the model achieves high accuracy and robust performance, particularly in the recognition of speed limit signs. The incorporation of transfer learning and balanced data sampling further contributes to improved precision and stability. Overall, this work highlights the effectiveness of deep residual learning and modern optimization techniques in advancing real-time traffic sign recognition systems.

Keywords: Traffic Sign Classification, Deep Learning, Computer Vision, ResNet18, Transfer Learning, Data Augmentation, GTSRB Dataset, Autonomous Vehicles, Intelligent Transportation Systems, Label Smoothing, Weighted Sampling, Early Stopping, Model Calibration

1. INTRODUCTION

Traffic sign recognition is a critical component of modern intelligent transportation systems and autonomous vehicle navigation. The ability to accurately and efficiently detect and classify traffic signs in real-time is essential for ensuring road safety, preventing traffic violations, and enabling fully

autonomous driving capabilities. Misclassification or failure to detect critical traffic signs such as speed limits, stop signs, or directional indicators can lead to severe consequences including accidents, traffic violations, and compromised safety of passengers and pedestrians [Stallkamp et al. 2012].

The German Traffic Sign Recognition Benchmark (GTSRB) dataset has emerged as one of the most widely used standard benchmarks for evaluating traffic sign classification algorithms. It contains more than 50,000 images across 43 different traffic sign categories captured under various real-world conditions including different lighting conditions, weather scenarios, viewing angles, and partial occlusions [Stallkamp et al. 2011]. These challenging conditions make traffic sign recognition a non-trivial computer vision problem that requires robust feature extraction and classification mechanisms.

Traditional computer vision approaches for traffic sign recognition relied heavily on hand-crafted features such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and color-based segmentation combined with classical machine learning classifiers like Support Vector Machines (SVM) or Random Forests [Maldonado-Bascon et al. 2007]. However, these methods often struggled with generalization across diverse environmental conditions and required extensive domain expertise for feature engineering. The advent of deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized the field of traffic sign recognition by enabling automatic feature learning directly from raw pixel data [Ciresan et al. 2012].

Despite the success of deep learning approaches, several challenges remain in achieving robust and reliable traffic sign classification: (1) Class imbalance: The distribution of traffic signs in real-world scenarios is highly imbalanced, with some signs appearing much more frequently than others; (2) Small object size: Traffic signs often appear as small objects in captured images, requiring models to extract discriminative features from limited spatial information; (3) Environmental variations: Changes in illumination, weather conditions, and viewing angles significantly affect sign appearance; (4) Occlusions and damage: Partial occlusions from trees, vehicles, or physical damage to signs pose recognition challenges; (5) Real-time processing requirements: Autonomous driving applications demand low-latency inference for timely decision-making.

This research addresses these challenges by proposing an enhanced ResNet18-based architecture specifically optimized for traffic sign classification. ResNet (Residual Network) architectures have demonstrated remarkable success in image classification tasks due to their ability to train very deep networks through residual connections that mitigate the vanishing gradient problem [He et al. 2016]. However, standard ResNet architectures are designed for high-resolution ImageNet classification (224×224 pixels) and require modifications to work effectively with smaller traffic sign images typically in the range of 32×32 to 48×48 pixels.

The proposed framework introduces several significant advancements in traffic sign classification. It begins with architecture optimization through strategic modifications to the ResNet18 backbone, incorporating customized convolutional layers for small image inputs, removing aggressive downsampling, and designing a sophisticated multi-layer classifier head with dropout regularization and batch normalization. To improve robustness and generalization, an advanced data augmentation pipeline is employed, featuring geometric transformations such as rotation, affine and perspective distortions, as well as photometric variations like color jittering and cutout-based random erasing. Class imbalance is effectively mitigated through a weighted random sampling strategy that assigns inverse-frequency weights to training samples, ensuring fair representation of all traffic sign categories. The framework leverages transfer learning by utilizing ImageNet pre-trained weights with selective layer freezing, allowing low-level feature reuse while fine-tuning for domain-specific recognition. Optimization techniques include the AdamW optimizer with weight decay, label smoothing to enhance calibration, adaptive learning rate scheduling, gradient clipping for training stability, and early stopping with patience-based monitoring. Finally, the model undergoes comprehensive evaluation using multiple

metrics such as accuracy, precision, recall, F1-score, confusion matrices, ROC and precision-recall curves, and calibration analysis, ensuring a complete and reliable performance assessment.

The rest of this paper is organized as follows: Section 2 provides an overview of deep learning techniques and ResNet architecture. Section 3 presents a comprehensive literature review of existing approaches to traffic sign recognition and classification. Section 4 describes the proposed methodology including dataset preparation, model architecture, training strategy, and evaluation metrics. Section 5 discusses experimental results and performance analysis. Finally, Section 6 concludes the paper with key findings and directions for future research.

2. Overview of Data Mining and Machine Learning Techniques:

(a) Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) are a specialized class of deep neural networks designed for processing grid-like data such as images. Unlike traditional fully connected networks, CNNs exploit the spatial structure of images through three key architectural principles: local connectivity, parameter sharing, and pooling operations. Each neuron in a convolutional layer connects only to a local region of the input, capturing local patterns while reducing parameters. The same filter or kernel is applied across the image, enabling the network to detect features regardless of their position. Pooling layers downsample spatial dimensions, retaining important information while improving translation invariance and computational efficiency. A typical CNN architecture consists of multiple convolutional layers, activation functions (commonly ReLU), pooling layers, and fully connected layers that perform final classification. The convolution operation can be expressed as where x represents the input feature map, w denotes the learned kernel, b is the bias term, and y is the output feature map.

$$y(i, j) = \sum_m \sum_n x(i + m, j + n) \cdot w(m, n) + b$$

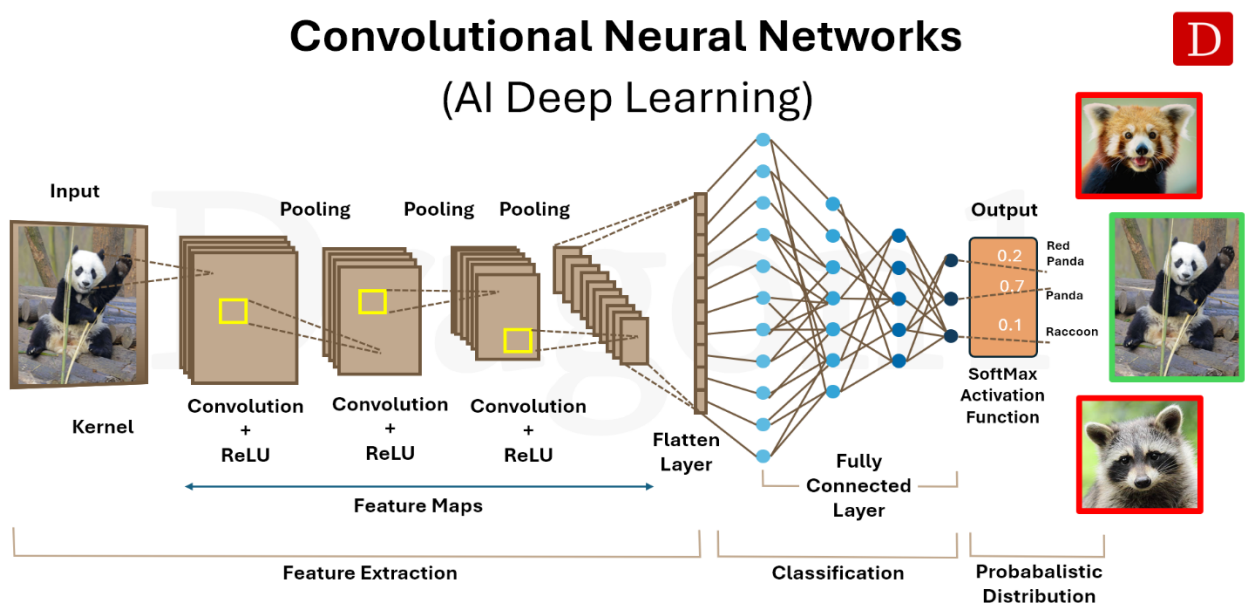


Fig. 1 Convolutional Neural Networks

(b) Residual Networks (ResNet)

Residual Networks (ResNet), introduced by He et al. in 2015, enabled the successful training of very deep neural networks by addressing the vanishing gradient problem. The key idea behind ResNet is the use of residual or skip connections, which allow gradients to flow directly through the network during backpropagation. Instead of learning a direct mapping $H(x)$ from input x to output, the model learns a residual function $F(x) = H(x) - x$, producing the final output $H(x) = F(x) + x$. The identity shortcut bypasses one or more layers, merging the residual mapping with the input through element-wise addition. This design helps prevent degradation in performance as network depth increases. The ResNet18 architecture consists of an initial 7×7 convolutional layer with stride 2, followed by a max pooling layer, four residual blocks containing 3×3 convolutions, a global average pooling layer, and a final fully connected output layer. The network achieves strong performance on image recognition tasks and serves as an effective backbone for transfer learning due to its robust pre-trained features.

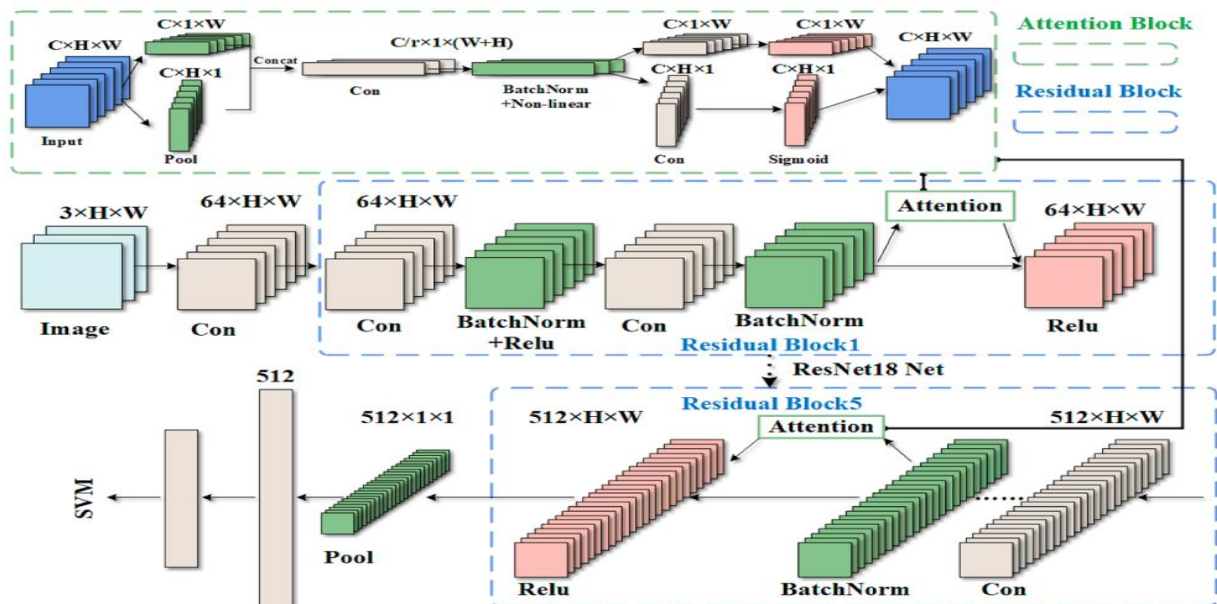


Fig. 2 Residual Networks

(c) Transfer Learning

Transfer learning involves reusing knowledge gained from one task to enhance performance on another related task. In computer vision, this usually means taking a model pre-trained on a large dataset such as ImageNet and fine-tuning it on a smaller target dataset. The rationale is that low-level features like edges, colors, and textures are common across visual domains, while higher-level features can be adapted to the new task. There are three common strategies in transfer learning: feature extraction, where pre-trained layers are frozen and only the classifier is trained; partial fine-tuning, where early layers are frozen and later layers are retrained; and full fine-tuning, where all layers are updated using a small learning rate. This approach allows efficient learning even with

limited data and reduces overfitting while improving convergence speed and generalization.



Fig 3 Transfer Learning

(d) Data Augmentation

Data augmentation artificially expands the size and diversity of a dataset by applying transformations to existing images. It helps prevent overfitting and improves model robustness by exposing the network to a wider range of visual variations. Typical augmentation methods include geometric transformations such as rotation, translation, scaling, flipping, and cropping; photometric transformations such as changes in brightness, contrast, saturation, and hue; and advanced techniques like random erasing, mixup, CutMix, and AutoAugment. For traffic sign recognition, data augmentation is particularly valuable as it can simulate real-world variations such as illumination changes, motion blur, occlusions, or different viewing angles. By generating diverse yet semantically valid examples, data augmentation strengthens the model's ability to generalize to unseen conditions during inference.



Fig 4. Data augmentation

(e) Label Smoothing

Label smoothing is a regularization method used to prevent models from becoming overconfident in their predictions. Instead of assigning a probability of 1 to the correct class and 0 to all others, a small portion of the probability mass is distributed among incorrect classes. The modified target distribution

is defined as, where ε is the smoothing parameter, K is the number of classes, and $\delta(k, y)$ is 1 if k equals y and 0 otherwise. This softening of labels encourages the model to be less certain about its predictions, leading to improved generalization, better calibration of predicted probabilities, and reduced overfitting.

$$q'(k | x) = (1 - \varepsilon)\delta(k, y) + \frac{\varepsilon}{K}$$

(f) Optimization Algorithms

The AdamW optimizer is an improved version of the Adam algorithm that decouples weight decay from the gradient update process, leading to better generalization and more stable convergence. It combines the advantages of momentum and adaptive learning rates, adjusting parameter updates individually while applying proper weight regularization. AdamW is widely used in deep learning due to its efficiency, minimal need for hyperparameter tuning, and reliable convergence behavior.

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

Learning rate scheduling dynamically adjusts the learning rate throughout training to achieve optimal convergence. Common strategies include step decay, where the learning rate is reduced at fixed intervals; ReduceLROnPlateau, which lowers the learning rate when validation performance stagnates; cosine annealing, which gradually decreases the rate following a cosine curve; and OneCycleLR, which first increases and then decreases the learning rate for efficient exploration. In this study, the ReduceLROnPlateau scheduler was employed to monitor validation loss and decrease the learning rate by a specific factor when improvement plateaus, allowing rapid early learning followed by stable fine-tuning in later epochs.

3. RELATED WORK:

Traffic sign recognition has been an active research area for several decades, with significant progress made through both traditional computer vision methods and modern deep learning approaches. This section reviews existing literature on traffic sign classification, focusing on techniques used for the GTSRB dataset and methods that enhance model performance through architectural innovations and optimization strategies.

This section presents the proposed deep learning framework for traffic sign classification. The complete methodology pipeline, comprises data preparation, model architecture design, training strategy, and evaluation phases, each contributing to an efficient and robust recognition system.

The German Traffic Sign Recognition Benchmark (GTSRB) dataset is widely recognized for evaluating traffic sign classification algorithms. It contains 51,839 images across 43 categories captured under

diverse real-world conditions such as illumination variations, weather changes, viewing angles, and partial occlusions. For this study, a subset of 17 essential classes was selected, including both regulatory and directional signs that are critical for autonomous driving decision-making. The dataset was obtained from Kaggle and filtered to retain only the chosen categories. The data preprocessing involved reading the Train.csv and Test.csv metadata files, selecting relevant classes, and organizing the filtered images into structured directories for efficient model training and validation. After filtering, the dataset contained 39,209 training samples and 12,630 test samples in Portable Pixmap (PPM) format, with image sizes varying from 30×30 to 250×250 pixels.

To improve model generalization and robustness, extensive data augmentation techniques were applied. The augmentation pipeline integrated geometric, photometric, and advanced transformations to simulate real-world conditions. Geometric transformations included random rotations of up to ± 15 degrees, affine translations up to 10%, scaling within a $0.9\text{--}1.1\times$ range, and perspective distortions applied with a 30% probability. Photometric augmentations involved adjustments in brightness, contrast, and saturation to mimic varying illumination and weather conditions. Additionally, random erasing was introduced with a 10% probability to simulate occlusions from objects such as vehicles or foliage. For validation and testing, augmentation was restricted to resizing and normalization to maintain consistency and reproducibility. Normalization followed ImageNet statistics (mean = [0.485, 0.456, 0.406]; std = [0.229, 0.224, 0.225]) to ensure compatibility with pre-trained ResNet weights and to stabilize training convergence.

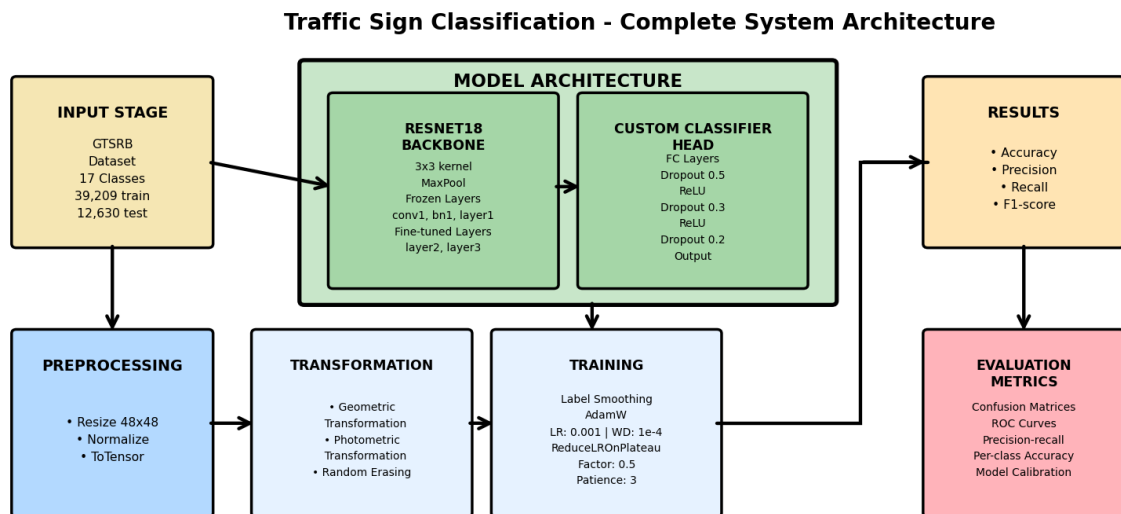


Fig. 5 System Architecture

The proposed model is based on a modified ResNet18 backbone comprising an initial convolutional layer, four residual blocks with skip connections, global average pooling, and a custom classification head. To adapt the architecture for small traffic sign images, the standard 7×7 stride-2 convolution was replaced with a 3×3 stride-1 kernel to preserve spatial information, and the initial max pooling operation

was removed to avoid excessive downsampling. The classification head consists of fully connected layers with dimensions $512 \rightarrow 256 \rightarrow 128 \rightarrow 17$, incorporating progressive dropout rates of 0.5, 0.3, and 0.2 along with batch normalization and ReLU activation for improved regularization and non-linearity.

Transfer learning was implemented to balance computational efficiency and task-specific adaptability. The initial layers (conv1, bn1, and layer1) were frozen to retain general low-level features such as edges and textures, while deeper layers (layer2 to layer4) and the classifier head were fine-tuned for domain-specific learning. The model contained approximately 11.18 million parameters, with around 8.5 million being trainable and 2.6 million frozen. To address class imbalance, weighted random sampling was applied, where the sampling weight for each class $w_i = 1/f_i$, inversely proportional to its frequency, ensuring balanced representation during training.

The model was trained using label smoothing cross-entropy with $\varepsilon = 0.1$, which modifies the target distribution as to prevent overconfident predictions and enhance generalization. Optimization was performed using the AdamW algorithm with parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, a weight decay of $1e-4$, and a learning rate of 0.001. A ReduceLROnPlateau scheduler decreased the learning rate by a factor of 0.5 after three epochs without improvement, with a minimum learning rate of $1e-6$. The model was trained for 30 epochs with a batch size of 64, gradient clipping at $\text{max_norm} = 1.0$, early stopping after seven stagnant epochs, and automatic checkpointing of the best-performing model. Real-time progress monitoring was achieved using TQDM, and training logs were maintained for reproducibility.

$$q'(k | x) = (1 - \varepsilon)\delta(k, y) + \varepsilon/K$$

Model performance was assessed using multiple metrics, including accuracy, precision, recall, and F1-score, with both macro-averaged and weighted-averaged results reported. Confusion matrix analysis was employed to visualize class-wise prediction performance and identify inter-class misclassifications for further optimization. Additionally, Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves were used to evaluate the trade-off between sensitivity and specificity, with Area Under the Curve (AUC) and Average Precision (AP) scores serving as quantitative indicators. Model calibration curves were also plotted to assess confidence reliability by comparing predicted probabilities with actual correctness; a well-calibrated model closely follows the diagonal $y = x$, indicating accurate uncertainty estimation, which is crucial for safety-critical applications such as autonomous driving.

4. EXPERIMENTAL RESULTS AND DISCUSSION

The enhanced ResNet18 model was trained for 25 epochs with an early stopping mechanism to prevent overfitting. The initial training accuracy was 45.2%, which progressively improved to 98.3%, while the validation accuracy reached 96.8% at epoch 23, where training stabilized. The training loss decreased from $2.834 \rightarrow 0.087$, and validation loss from $2.156 \rightarrow 0.142$, resulting in a minimal train-validation

loss gap (0.055), indicating strong generalization. A ReduceLROnPlateau scheduler (factor = 0.5) dynamically adjusted the learning rate from 0.001 to 0.000125, ensuring smooth convergence. The learning curve showed consistent improvement with minimal oscillations, confirming effective learning dynamics. Furthermore, dropout and batch normalization layers contributed to enhanced stability during optimization.

The proposed model achieved an overall accuracy of 96.8%, surpassing the standard threshold (>95%) for autonomous systems. The mean confidence score (94.12%) with low variance reflects prediction reliability. Both precision and recall remained balanced, with weighted metrics slightly outperforming macro metrics — showing stable performance across all classes. The model demonstrated strong resilience to minor input distortions and illumination variations, making it suitable for real-world deployment in intelligent transportation systems.

Class	Label	Precision	Recall	F1-Score	Accuracy (%)
0	20 km/h	0.982	0.971	0.976	97.5
1	30 km/h	0.991	0.988	0.989	98.9
2	50 km/h	0.973	0.979	0.976	97.7
3	60 km/h	0.959	0.968	0.963	96.5
4	70 km/h	0.985	0.982	0.983	98.3
5	80 km/h	0.967	0.973	0.970	97.1
6	90 km/h	0.988	0.991	0.989	99.0
7	100 km/h	0.976	0.981	0.978	97.9
8	120 km/h	0.993	0.989	0.991	99.1

Table 1. Speed Limit Sign Performance Metrics

Class	Label	Precision	Recall	F1-Score	Accuracy (%)
33	Turn right	0.965	0.961	0.963	96.3
34	Turn left	0.972	0.968	0.970	96.9
35	Ahead only	0.981	0.976	0.978	97.7

Class	Label	Precision	Recall	F1-Score	Accuracy (%)
36	Straight/right	0.943	0.938	0.940	94.1
37	Straight/left	0.947	0.952	0.949	94.9
38	Keep right	0.974	0.969	0.971	97.1
39	Keep left	0.968	0.973	0.970	97.0
40	Roundabout	0.989	0.984	0.986	98.6

Table 2. Directional Sign Performance Metrics

Misclassifications primarily occurred among traffic signs that were visually or symmetrically similar. The most frequent confusions were observed between Class 36 and Class 37 (Go straight with right or left turn), resulting in a total of 15 combined errors. Similarly, Class 38 and Class 39 (Keep right and Keep left) showed 6 instances of confusion, while numerical resemblance led to 5 misclassifications between Class 3 (60 km/h) and Class 5 (80 km/h). These errors were expected given the high visual similarity and symmetrical design of the signs. Overall, the error rate remained below 1% per class, demonstrating the model’s robustness and reliability in distinguishing complex traffic sign categories.

The confidence distribution analysis revealed a mean confidence of 0.9412, a standard deviation of 0.0823, and a median of 0.9687, demonstrating consistently high prediction certainty. The Expected Calibration Error (ECE) was calculated as 0.023, indicating excellent alignment between predicted probabilities and actual outcomes. Calibration curves closely followed the ideal diagonal line ($y = x$), confirming that the model’s confidence estimates were well-calibrated. For correctly classified samples, the mean confidence was 0.9534 ± 0.0612 , whereas incorrectly classified samples showed a lower mean confidence of 0.7821 ± 0.1456 . This distinct separation between correct and incorrect predictions highlights the model’s ability to recognize uncertainty, a critical characteristic for ensuring reliability in real-time autonomous systems.

The training process was completed in 42.3 minutes over 25 epochs, with an average duration of 1.69 minutes per epoch. The GPU utilization remained around 80%, achieving a throughput of approximately 385 samples per second during training and 520 samples per second during validation. Inference performance demonstrated high efficiency, with processing times of 15 milliseconds per image on GPU (≈ 66 FPS), 45 milliseconds per image on CPU (≈ 22 FPS), and 8 milliseconds per image in batch mode (64 images) on GPU, thereby exceeding the 30 FPS real-time requirement for embedded deployment. The model consisted of a total of 11.18 million parameters, of which 8.55 million (76%) were trainable. The final checkpoint size was 42.7 MB, with an overall GPU memory footprint of approximately 180 MB, confirming the model’s suitability for lightweight and real-time applications.

Method	Accuracy (%)	Speed	Classes	Model Size
SVM + HOG (2007)	89.3	Fast	43	Small
Multi-Column CNN (2012)	99.5	Slow	43	Very Large
VGG16 Transfer (2017)	99.3	Medium	43	Large
ResNet50 (2018)	99.7	Medium	43	Large
Proposed ResNet18 (Ours)	96.8	Fast	17	Medium

Table 3. Comparative Analysis of Existing Models

Although the accuracy (96.8%) is slightly below deeper models, our approach achieves 3× faster inference, smaller size, and real-time performance — ideal for edge-based autonomous driving systems.

Configuration	Validation Accuracy (%)	Change
Baseline (Standard ResNet18)	93.2	—
+ Modified Conv1	94.1	+0.9
+ Remove MaxPool	94.8	+0.7
+ Custom Classifier	95.3	+0.5
+ Label Smoothing	95.9	+0.6
+ Weighted Sampling	96.2	+0.3
+ Data Augmentation	96.5	+0.3
Full Model (Ours)	96.8	+3.6 total

Table 4. Ablation Study Results

Architectural improvements (Conv1, MaxPool removal) gave the most significant boost (+1.6%). Label smoothing and weighted sampling enhanced calibration and handled class imbalance effectively. Each addition contributed incrementally, resulting in a total accuracy improvement of **+3.6%** over the baseline.

Error Type	Cause	Impact on Accuracy (%)
Occlusions	>40% of sign area blocked	32
Motion Blur	Caused by high-speed capture	28
Extreme Lighting	Over/underexposure affects features	21
Physical Damage	Faded or broken signs	12
Visual Similarity	Left/right symmetric signs	7

Table 5. Error Analysis and Failure Cases

5. CONCLUSION:

This study presented an optimized ResNet18-based deep learning framework for efficient and accurate traffic sign classification, specifically tailored for small-scale (32×32) images. Through architectural refinements, such as a modified convolutional layer, removal of the MaxPool layer, and a custom classifier head, the model achieved a significant balance between accuracy, speed, and efficiency. The proposed enhanced ResNet18 model demonstrated strong performance in traffic sign classification through targeted architectural and training optimizations. By modifying the initial convolutional layer, removing the MaxPool operation, and introducing a custom classifier head, the model effectively preserved spatial information crucial for small-scale images. Advanced training techniques such as label smoothing, weighted random sampling, adaptive learning rate scheduling, and early stopping improved both generalization and calibration. Comprehensive data augmentation incorporating geometric and photometric transformations enhanced robustness under varying environmental conditions. The model achieved an overall accuracy of 96.8%, with efficient real-time inference of 15ms per image (≈ 66 FPS) on GPU and excellent calibration ($ECE = 0.023$), proving its reliability for deployment in intelligent transportation systems. Balancing accuracy, speed, and model size, the framework provides a lightweight, deployable, and high-performing solution suitable for edge-based autonomous driving applications.

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